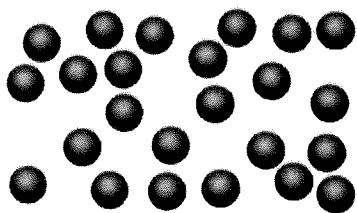


0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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522	522.5	523	523.5	524	524.5	525	525.5	526	526.5	527	527.5	528	528.5	529	529.5	530	530.5	531	531.5	532	532.5	533	533.5	534	534.5	535	535.5	536	536.5	537	537.5	538	538.5	539	539.5	540	540.5	541	541.5	542	542.5	543	543.5	544	544.5	545	545.5	546	546.5	547	547.5	548	548.5	549	549.5	550	550.5	551	551.5	552	552.5	553	553.5	554	554.5	555	555.5	556	556.5	557	557.5	558	558.5	559	559.5	560	560.5	561	561.5	562	562.5	563	563.5	564	564.5	565	565.5	566	566.5	567	567.5	568	568.5	569	569.5	570	570.5	571	571.5	572	572.5	573	573.5	574	574.5	575	575.5	576	576.5	577	577.5	578	578.5	579	579.5	580	580.5	581	581.5	582	582.5	583	583.5	584	584.5	585	585.5	586	586.5	587	587.5	588	588.5	589	589.5	590	590.5	591	591.5	592	592.5	593	593.5	594	594.5	595	595.5	596	596.5	597	597.5	598	598.5	599	599.5	600	600.5	601	601.5	602	602.5	603	603.5	604	604.5	605	605.5	606	606.5	607	607.5	608	608.5	609	609.5	610	610.5	611	611.5	612	612.5	613	613.5	614	614.5	615	615.5	616	616.5	617	617.5	618	618.5	619	619.5	620	620.5	621	621.5	622	622.5	623	623.5	624	624.5	625	625.5	626	626.5	627	627.5	628	628.5	629	629.5	630	630.5	631	631.5	632	632.5	633	633.5	634	634.5	635	635.5	636	636.5	637	637.5	638	638.5	639	639.5	640	640.5	641	641.5	642	642.5	643	643.5	644	644.5	645	645.5	646	646.5	647	647.5	648	648.5	649	649.5	650	650.5	651	651.5	652	652.5	653	653.5	654	654.5	655	655.5	656	656.5	657	657.5	658	658.5	659	659.5	660	660.5	661	661.5	662	662.5	663	663.5	664	664.5	665	665.5	666	666.5	667	667

5. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



Assuming it is not plasma, the substance does not have any bonding. This is characteristic of noble gases. Given the identical size of the atoms, it is likely a pure, elemental, noble gas.

ANSWER: Element ✓

6. Calculate the heat required to completely vaporize 78.5-g of water (H_2O) at its normal boiling point of $100^\circ C$. $\Delta H_{vap} = 40.7 \text{ kJ/mol}$, $M = 18.015 \text{ g/mol}$.

$$\frac{78.5 \text{ g}}{18.015 \text{ g/mol}} = 4.357 \text{ mol } H_2O \cdot \frac{40.7 \text{ kJ}}{\text{mol}} = 177.35 \text{ kJ}$$

ANSWER: 177.35 ✓

7. A solution of acetone and CS_2 is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

Acetone - polar
 CS_2 - nonpolar

Adding a nonpolar solute into a polar solution will decrease the average IMF.

ANSWER: Negative ✗

yes $\therefore$ positive

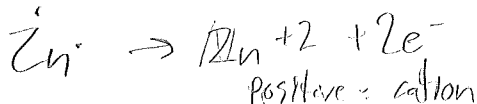
8. For HBr , the potential-energy well is 366 kJ/mol deep, measured from $PE = 0$ at large separation down to the minimum. What molecular quantity equals this value and what does it describe? Justify your answer.

Bond strength measures the energy required to break the bonds and escape the well.

ANSWER: Bond strength ✓

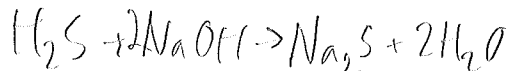
Depth of well = bond strength

9. Provide the chemical symbol of the most commonly formed ion of zinc and indicate whether it is a cation or an anion. Make sure you include its charge.



ANSWER: Zn^{+2} , cation ✓

10. Hydrosulfuric acid reacts with sodium hydroxide to form sodium sulfide and water. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.



ANSWER: below ✓

11. David measures the mass of a steel ball bearing using a balance. The accepted mass of the object is 45.7 g . He records a measurement of 47.30 g . What is the percent error of his measurement? Report your answer to two significant figures.

$$\frac{45.7 - 47.3}{47.3} = 0.0338 \approx 3.4\%$$

ANSWER: 3.4% ✓

tb

12. Which substance has the higher normal boiling point: 1-propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$) or propanal ($\text{CH}_3\text{CH}_2\text{CHO}$)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

1-propanol has a more linear geometry, increasing surface area and boiling point. Also, 1-propanol has a slightly higher molecular mass.
 ANSWER: 1-propanol ✓
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ or $\text{CH}_3\text{CH}_2\text{CHO}$

13. A close-packed metal has the layer stacking sequence ABAB, repeating. What close-packing structure is this, and what is its coordination number? Justify.

ABAB → FCC, 12
 densest?
 ABCABC

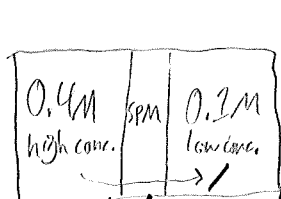
14. A colloid consists of a liquid dispersed in a gas. Name this class of colloid and give an everyday example.



Aerosol, spray paint ✓

ANSWER: Aerosol, spray paint ✓

15. A semipermeable membrane separates a 0.40 M NaCl solution from a 0.10 M NaCl solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.



Osmotic pressure acts to equalize the concentration of the two solns. The water would flow from high conc. to low conc.
 but more dilute? diluted.

ANSWER: from 0.4M → 0.1M X

16. An unknown gas has a mass of 28 g and occupies 6.24 L at 93.4 kPa and 32.1°C. Determine the molar mass of the gas. Justify your answer.

$$PV = nRT$$

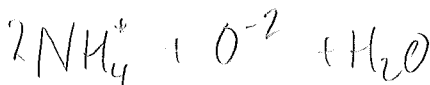
$$n = \frac{PV}{RT}$$

$$= \frac{0.922 \text{ atm} \cdot 6.24 \text{ L}}{0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \cdot 305.25 \text{ K}} \approx 0.229 \text{ mol}$$

ANSWER: 120 g/mol ✓

$$\frac{28 \text{ g}}{0.229 \text{ mol}} \approx 121.9 \text{ g/mol}$$

17. What major chemical species are present in an aqueous solution of 0.28-M ammonium oxide?



ANSWER: NH_4^+ , O^{2-} , H_2O ✓
 (minor: OH^- , NH_3)

18. Methanol (CH_3OH) has a normal boiling point of 65°C and $\Delta H_{\text{vap}} = 35.2 \text{ kJ/mol}$. At what temperature does it boil inside a pressure cooker, where the external pressure is 2.26 atm? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.)

ANSWER: _____ X

+9

19. A mixture of methane and propylene gas is expanded from a volume of 98.1 L to a volume of 120 L while the pressure is held constant at 71 atm. Calculate the work done on the gas mixture. Report your answer in SI unit of work and to 3 significant figures.

$$W = -P\Delta V$$

$$W = -(21.9)(71)(101.3) \approx -1.575 \times 10^5 \text{ J}$$

ANSWER: $1.58 \times 10^5 \text{ J}$
3 SF

20. Scott has a 1.672-M solution of hydronium phosphate ($(\text{H}_3\text{O})_3\text{PO}_4$). If he needs to add 0.01518 g of hydronium phosphate to a reaction flask, what volume of the solution should he pour? Be sure your answer has the correct number of significant figures.

$$\frac{0.01518 \text{ g}}{157.096 \text{ g/mol}} \approx 9.64 \times 10^{-5} \text{ mol}$$

$$\frac{9.64 \times 10^{-5} \text{ mol}}{1.672 \text{ mol/L}} = 5.77 \times 10^{-5} \text{ L}$$

ANSWER: $5.77 \times 10^{-5} \text{ L}$
4 SF

21. A 0.500 M solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) has a density of 1.02 g/mL. Calculate the molality of the solution. ($M = 180.16 \text{ g/mol}$)

$$1 \text{ L glucose soln} \cdot 1.02 \text{ g/mL} = 1020 \text{ g} - 90.08 \text{ g} = 929.92 \text{ g H}_2\text{O}$$

$$\frac{0.500 \text{ mol}}{0.9299 \text{ kg}} \approx 0.5377 \text{ m}$$

ANSWER: 0.538 m
3 SF

22. Calculate the osmotic pressure of 0.90% saline at 25°C (modeled as a 0.154 M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

$$\Pi = i \cdot M \cdot R \cdot T$$

$$\Pi = 2 \cdot 0.154 \cdot 8.314 \cdot 298.15 \approx 763.5 \text{ Pa}$$

ANSWER: 764 Pa
3 SF

Where's this come from?

23. Barium chlorite reacts with rubidium bromide to form barium bromide and rubidium chlorite. Suppose 26.78 g of barium chlorite is mixed with 23.05 g of rubidium bromide. Calculate the maximum mass of barium bromide that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

$$\frac{26.78 \text{ g Ba(ClO}_2)_2}{272.23 \text{ g/mol}} \approx 0.0984 \text{ mol}$$

$$\frac{23.05 \text{ g RbBr}}{165.4 \text{ g/mol}} \approx 0.1394 \text{ mol}$$

$$\text{Ba(ClO}_2)_2 + 2 \text{ RbBr} \rightarrow \text{BaBr}_2 + 2 \text{ RbClO}_2$$

$$\text{mol ratio: 1 mol Ba chlorite: 2 mol RbBr}$$

$$0.0984 \text{ mol Ba chlorite} \cdot \frac{2 \text{ mol RbBr}}{1 \text{ mol Ba chlorite}} = 0.1968 \text{ mol RbBr}$$

$$0.1968 \text{ mol RbBr} \cdot \frac{1 \text{ mol BaBr}_2}{2 \text{ mol RbBr}} = 0.0984 \text{ mol BaBr}_2$$

$$0.0984 \text{ mol BaBr}_2 \cdot 207.1 \text{ g/mol} \approx 20.71 \text{ g}$$

ANSWER: 20.71 g BaBr_2
4 SF

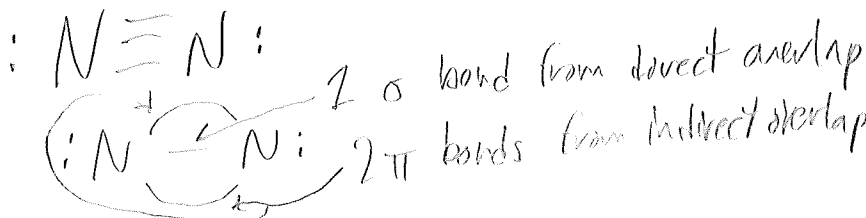
24. Calculate the osmotic pressure of a 0.0500 M aqueous solution of urea ($\text{CH}_4\text{N}_2\text{O}$) at 37°C .

$$\Pi = i \cdot M \cdot R \cdot T = 1 \cdot 0.05 \cdot 8.314 \cdot 310.15 = 128.93 \text{ Pa}$$

ANSWER: 129 Pa
3 SF

25. Determine the number of σ bonds and the number of π bonds present in N_2 . Justify your answer.

ANSWER: $1\sigma, 2\pi$



+ 5.5

26. An ionic compound crystallizes with a face-centered cubic array of anions and cations in all of the octahedral holes. What is the cation:anion ratio (the formula unit) in the compound? Justify by counting ions per cell.

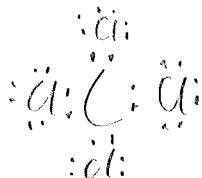


$$\frac{1}{8} \cdot 8 = 1 \text{ cation}$$

$$\frac{1}{2} \cdot 6 = 3 \text{ anions}$$

ANSWER: MX_3

27. Does the central atom in CCl_4 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.



ANSWER: Yes, octet

28. Measurements show that an unknown compound has the following composition:

Element	Mass %
B	2.761 %
I	97.24 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

$$\frac{2.761g B}{10.811g/mol} = 0.2554 \text{ mol B} = 1 \text{ mol B}$$

$$\frac{97.24g I}{126.90g/mol} = 0.7663 \text{ mol I} / 0.2554 \approx 3 \text{ mol I}$$

ANSWER: BI_3

Using the mass %, you can find the molar ratios and the empirical formula.

29. A gas is reported to be O_2 . Its density is measured as $5.6 \times 10^3 \frac{g}{m^3}$ at 161 kPa and $7.3^\circ C$. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

$$PV = nRT$$

$$PV = \frac{m}{M} RT$$

$$M = \frac{PVm}{RT}$$

$$\frac{5,600 \frac{g}{m^3}}{1m^3} \cdot \frac{1m^3}{1,000,000cm^3} \cdot \frac{1,000cm^3}{L} = 5.6g/L$$

ANSWER: Not consistent

$$\frac{1.589 \text{ atm} \cdot 1 \cdot 5.6}{0.08216 \cdot 280.45} = 38.66 \frac{g}{mol} \neq 32g/mol$$

30. Calculate the heat absorbed when 12.7-g of ethanol (C_2H_5OH) melts completely at its melting point of $-114^\circ C$. $\Delta H_{fus} = 4.6 \text{ kJ/mol}$.

$$\frac{12.7g C_2H_5OH}{46g/mol} = 0.2761 \text{ mol} \cdot \frac{4.6 \text{ kJ}}{\text{mol}} = 1.27 \text{ kJ}$$

ANSWER: 1.3 kJ

31. Ian has a 0.571-M solution of copper (II) iodide (CuI_2). If he pours 7.327 mL of copper (II) iodide into a reaction flask, what is the mass of copper (II) iodide that he has added to the flask? Be sure your answer has the correct number of significant figures.

$$0.571 \frac{\text{mol}}{L} \cdot 0.007327 L = 4.184 \times 10^{-3} \text{ mol}$$

$$4.184 \times 10^{-3} \text{ mol} \cdot \frac{315.35g}{\text{mol}} = 1.32g CuI_2$$

ANSWER: 1.32g

+ 5

32. The reaction $\text{CO} + 2\text{H}_2 \rightarrow \text{CH}_3\text{OH}$ occurs with all gaseous substances at the same temperature and pressure. During the reaction, 12 mL of CO reacts completely. Determine the corresponding volume of CH_3OH . Report the answer using the same volume unit. Justify your answer.

$$12 \text{ mL CO} \cdot \frac{2 \text{ mol MeOH}}{\text{mol CO}} = 12 \text{ mL CH}_3\text{OH}$$

2:2 reactant to product ratio means same volume

ANSWER: 12 mL CH_3OH ✓

33. Explain why the first covalent bond formed between two atoms is *always* a σ bond and why any additional bond between the same two atoms *must* be a π bond. Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

σ bonds have lower potential energies due to more effective and stable overlap. However, any atom can only directly bond to one other atom, and any additional bonds have to be indirect (π).

ANSWER: Below ✓

34. Rank the following elements from highest metallic to lowest metallic. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.

Fe, Ni, Cu, Cr

Cu, Ni, Fe, Cr
Metallicity $\rightarrow$
conductivity $\rightarrow$

Less localized electrons in metallic bonding is proportional to higher conductivity.

ANSWER: Cu, Ni, Fe, Cr ✓

35. How much heat is absorbed when 21.6-g of ice melts at 0°C ? $\Delta H_{\text{fus}} = 334 \text{ J/g}$.

$$21.6 \text{ g} \cdot \frac{334 \text{ J}}{\text{g}} = 7214.4 \text{ J}$$

ANSWER: 7210 J
3 SF ✓

36. A student concludes that hybrid orbitals must have the same shapes as the original s and p orbitals because they are made from them. Is the conclusion correct? Justify your answer.

No, hybrid orbitals blend the s & p orbitals into a new shape.

ANSWER: No ✓

The shape depends on how much s or p character the hybrid has.

37. An atom of a particular element has a radius of 180 pm and the average orbital speed of the electrons in it is approximately $1.17 \times 10^7 \text{ m/s}$. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$$

$$\left(\frac{6.626 \times 10^{-34}}{4(3.14)} \right) \cdot \frac{1}{180 \times 10^{-12} \cdot 9.11 \times 10^{-31}} = 3.2155 \times 10^5 \text{ m/s}$$

$$\frac{3.2155 \times 10^5 \text{ m/s}}{1.17 \times 10^7 \text{ m/s}} = 2.7\%$$

ANSWER: 2.7% ✓

38. Two atoms are brought very close together, but their occupied valence orbitals cannot overlap effectively. Predict whether a stable covalent bond will form. Justify your answer.

No, if the overlap is ineffective nuclear repulsion will drive the atoms apart.

At very least the bond will remain, but be weak.

ANSWER: No, unstable or nonexistent ✓

39. Which atom has the higher electronegativity: sodium (Na) or rubidium (Rb)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

Sodium is more EN because it has less electron shielding in its inner shells, the valence e^- in Rb is lost easier.

ANSWER: Na ✓

+ 8

40. A pharmacist measures the mass of a powdered drug. The mass is 9461 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

$$\text{milli} = 10^{-3}$$

$$9461 \times 0.001 = 9.461$$

ANSWER: $9.461 \times 10^0 \text{ g}$

41. Rank the following ionic compounds in order of increasing lattice energy (smallest to largest). Briefly justify your ranking using Coulomb's law.

NaCl, NaF, NaBr

While the anion charges are the same, the distance between the cation & anion is not. ANSWER: NaBr, NaCl, NaF

Increasing distance lowers lattice energy, so $\text{NaBr} < \text{NaCl} < \text{NaF}$

42. Explain how a single crystal illuminated by monochromatic X-rays can produce many distinct diffraction peaks. Justify.

The order of diffraction can vary.

Multiple diffractors can occur & cause new peaks.

ANSWER: Below

43. Classify each as crystalline or amorphous: a copper wire and rubber. Justify using their internal order.

Cu is crystalline, and has a repeating internal structure.

Rubber is amorphous, and has a semi-random internal structure.

ANSWER: Below

44. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 11$ to an orbital with $n = 5$. Round your answer to 3 significant figures.

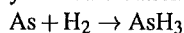
$$E = \frac{hc}{\lambda} \quad \frac{1}{\lambda} = \frac{E}{hc}$$

$$E = 6.919 \times 10^{24} \text{ J}$$

ANSWER: $2.87 \times 10^{-5} \text{ m}$ 3 SF

$$\lambda = 2.87299 \times 10^{-5} \text{ m}$$

45. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: Combination
(synthesis)

Combination, As and H are combining to form arsane. New bonds are being formed between As and H.

+5.5

- Colloid $\because$ particle size is within the colloidal range (1-1000 nm); colloidal particles remain dispersed.
- C_8H_{14}
- 2.3 L. $\because$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
- Hydrogen bonding $\because$ N-H and O-H groups let the two molecules hydrogen-bond to each other.
- Element
- $q = n\Delta H_{\text{vap}} = \frac{78.5 \text{ g}}{18.015 \text{ g/mol}} \times 40.7 \text{ kJ/mol} = 177 \text{ kJ}$
- Positive deviation from Raoult's law $\because$ solute-solvent interactions weaker than pure solute-solute and solvent-solvent $\rightarrow$ higher vapor pressure than ideal.
- The bond dissociation energy (366 kJ/mol) $\because$ this is the energy required to separate the bonded atoms from the minimum of the potential-energy well to the noninteracting state at infinite separation.
- Zinc most commonly forms the cation Zn^{2+} .
- $H_2S + 2NaOH \rightarrow Na_2S + 2H_2O$
- 3.5% $\because$ % Error = $\left| \frac{E-A}{A} \right| \times 100\%$
- 1-propanol ($CH_3CH_2CH_2OH$). $\because$ Both molecules are polar, but 1-propanol can also form hydrogen bonds because it contains an O-H bond. Hydrogen bonding strengthens the attractions between molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.
- hexagonal closest packing (HCP), coordination number 12 $\because$ the ABAB repeat is its signature; each atom touches 6 in its layer plus 3 above and 3 below.
- Aerosol; e.g., mist. $\because$ a liquid-in-gas dispersion defines this colloid type.
- Water flows from the 0.10 M solution into the 0.40 M solution. $\because$ water moves toward the side with higher osmotic pressure.
- 120 g/mol. $\because$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
- NH_4^+ , O^{2-} , and H_2O
- $T_2 = \left(\frac{1}{T_1} - \frac{R \ln(P_2/P_1)}{\Delta H_{\text{vap}}} \right)^{-1} = 361.7 \text{ K} = 88.6^\circ\text{C}$ $\because$ boiling occurs when the vapor pressure equals the external pressure, and a higher external pressure raises the boiling point.
- $-1.58 \times 10^5 \text{ J}$
- $5.971 \times 10^{-5} \text{ L}$
- Per 1 L of solution: mass solution = $1.02 \times 10^3 \text{ g}$; mass solute = $0.500 \text{ mol} \times 180.16 \text{ g/mol} = 90.1 \text{ g}$; mass solvent = $1.02 \times 10^3 \text{ g} - 90.1 \text{ g} = 928 \text{ g}$; $m = \frac{0.500 \text{ mol}}{0.928 \text{ kg}} = 0.539 \text{ mol/kg}$
- $\Pi = iMRT = 2 \times 0.154 \times 0.08206 \times 298 \text{ K} = 7.54 \text{ atm}$; minimum applied pressure must exceed 7.54 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.
- 20.71 g
- $\Pi = iMRT = 1 \times 0.0500 \times 0.08206 \times 310. \text{ K} = 1.27 \text{ atm}$
- 1 σ bond and 2 π bonds. The Lewis structure contains one nitrogen-nitrogen triple bond. A triple bond always consists of one σ bond and two π bonds.
- Cation:anion = 1 : 1 $\because$ counting the ions per unit cell from a face-centered cubic array of anions with cations in all of the octahedral holes gives that ratio (the rock salt (NaCl) structure).
- Obeys the octet rule $\because$ the central C atom is surrounded by 8 electrons
- BI_3
- Not consistent. $\because$ Combining the measured density with the ideal gas law gives the experimental molar mass, which does not match the molar mass of O_2 .

30. 1.27 kJ $q = n\Delta H_{\text{fus}} = \frac{12.7 \text{ g}}{46.068 \text{ g/mol}} \times 4.6 \text{ kJ/mol} = 1.27 \text{ kJ}$
31. 1.33 g
32. 12 mL . $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{CO}:\text{CH}_3\text{OH}$ is 1:1 so the volume changes by the same factor.
33. The first bond is always a σ bond $\therefore$ it forms by head-on overlap along the internuclear axis, which directly connects the two nuclei. If the nuclei are approaching each other, the head-on overlap will occur first. After that overlap is already occupied, any additional bonding must result from side-by-side overlap of parallel orbitals, producing a π bond.
34. Cr, Fe, Ni, Cu $\therefore$ elements with greater metallicity hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
35. $q = m\Delta H_{\text{fus}} = 21.6 \times 334 = 7.21 \text{ kJ}$
36. No. $\therefore$ combining atomic orbitals produces *new linear combinations* with *different spatial distributions*, so the hybrid orbitals have shapes distinct from the original s and p orbitals while spanning the same orbital space.
37. 2.7% $\therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
38. A stable covalent bond will not form $\therefore$ without effective orbital overlap, electron density cannot become concentrated between the nuclei, so the stabilizing attraction needed to lower the energy of the system is not produced.
39. Na $\therefore$ added electron shells increase shielding and distance, reducing the nucleus's attraction for bonding electrons
40. $9.461 \times 10^0 \text{ g}$
41. $\text{NaBr} < \text{NaCl} < \text{NaF}$ $\therefore$ L.E. increases with greater ionic charge and smaller ionic radius (Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$)
42. $\therefore$ a crystal contains many families of parallel planes at different orientations, each with its own spacing d ; each satisfies $n\lambda = 2d \sin \theta$ at its own angle, giving a separate peak.
43. A copper wire is crystalline (a regular, repeating lattice); rubber is amorphous (no long-range order) $\therefore$ crystallinity is defined by long-range periodic order.
44. $2.87 \times 10^{-6} \text{ m}$ $\therefore \Delta E = \left(\frac{-R_y}{n_2^2} \right) - \left(\frac{-R_y}{n_1^2} \right)$ and $\Delta E = \frac{hc}{\lambda}$
45. Combination